Technical Project Log: Cessna 210 Airframe Reconstruction

NASA Variant Parametric Modeling & Mesh Generation

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1 Project Introduction

This log documents the 3D reconstruction and numerical discretization of a Cessna 210 Centurion airframe using NASA's OpenVSP (Vehicle Sketch Pad). The primary goal was to translate 2D engineering blueprints into a watertight 3D parametric model suitable for future avionics and telemetry integration within the Team Antarez framework.

2 Geometric Modeling & Parameter Verification

The airframe was constructed using a multi-component assembly. A significant portion of the workflow involved the cross-verification of provided instructional dimensions against original 3-view drawings.

Component	**Finalized Dimension**	**Status/Notes**
Main Wingspan	36.748 ft	Verified (NACA 2412)
Fuselage Length	24.580 ft	Scaled to Reference
Main Wing Root Chord	5.525 ft	Parameterized
Horizontal Tail Span	12.990 ft	Verified
Vertical Tail Height	**7.10 ft**	**Corrected from 2.94 ft typo**

Table 1: Finalized Design Parameters for Cessna 210 Model.

2.1 Correction of Specification Errors

During the construction of the empennage, a major discrepancy was identified in the source instructions, which listed the vertical stabilizer height at 2.94 ft. Through visual ratio analysis of the NASA blueprints, this was identified as a scaling typo and corrected to ∼7.10 ft to maintain aerodynamic stability and geometric fidelity.

3 Numerical Mesh Generation

The model was discretized into a Vortex Lattice Method (VLM) mesh to analyze lift-induced pressure gradients.

- **Discretization:** A total of 5,731 nodes were generated to ensure surface continuity.
- **Wing Geometry:** Implemented 2.0° dihedral for lateral stability.
- **Solver Analysis:** Initial VSPAERO trials were conducted across an Angle of Attack (α) range of 0° to 5°.

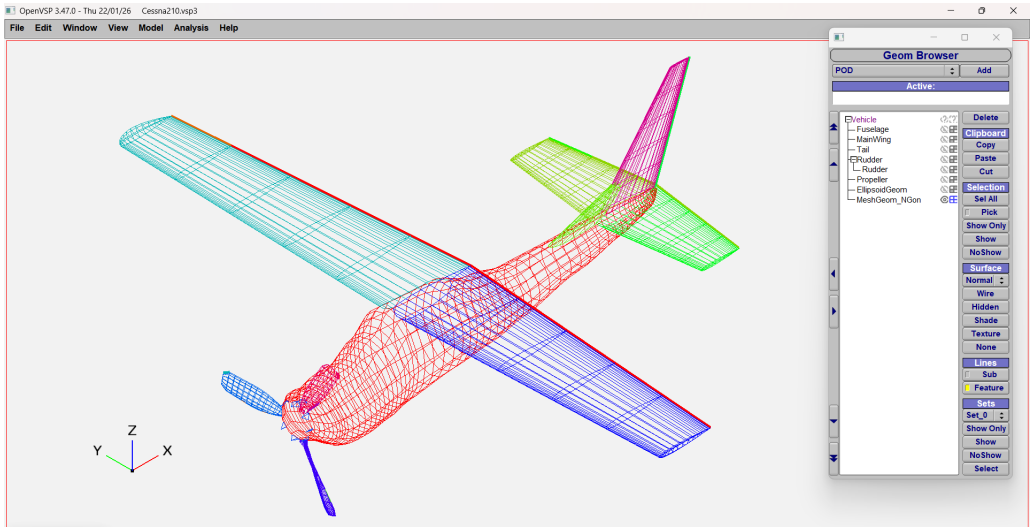

Figure 1: Finalized VLM Mesh showing surface discretization and node density.

4 Computational Post-Mortem

Numerical analysis encountered a Kutta condition detection failure ($N_k = 0$) during high-density solver iterations. This was diagnosed as a byproduct of excessive Num-U discretization in the trailing-edge geometry. The project successfully achieved its goal of creating a "simulation-ready" parametric model, with geometric watertightness verified through mesh visualization.

5 Conclusion

The project successfully bridge the gap between theoretical aircraft geometry and computational modeling. The finalized `.vsp3` file is archived for subsequent aerodynamic stability testing and electronic hardware-in-the-loop (HIL) simulation.